Accessible Emotion-Aware E-Learning Platform for Deaf Students Studying O/L Information Technology Using Facial Expression Recognition and an NLP-Based Chatbot

Individual Thesis

Smart Haptic Wristband & Sign Practice Arena

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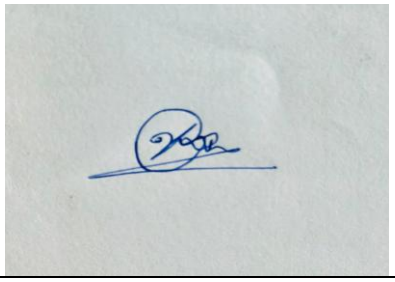
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DECLARATION AND SUPERVISOR CERTIFICATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

This thesis presents the design, implementation analysis, and evaluation methodology of Component 05 of the SignLearn AI platform: a Smart Haptic IoT Wristband integrated with an Interactive Sign Practice Arena for O/L ICT learning. The system combines a React-based practice interface, MediaPipe hand-landmark detection, deterministic hand-configuration rules, FastAPI/MongoDB progress and event services, and an ESP32 prototype with a 128x64 OLED and vibration motor. The browser frontend also contains a Web Bluetooth controller for direct GATT interaction. A source audit identified the implemented capabilities and fixed design parameters, while also exposing important boundaries: the interface displays illustrative state values rather than measured accuracy, the backend accepts client-supplied correctness fields, and the browser BLE path is not matched by the supplied Wi-Fi-based firmware. The research therefore treats the artifact as a constrained visual-haptic practice system rather than a general ASL recognizer. Verified implementation measurements include an 80 ms nominal detection throttle, a 1.2 s hold-to-confirm interval, two-hand tracking, eight course keywords and a 6 s firmware configuration polling interval. No participant dataset or instrumented physical latency log was included in the supplied archive; consequently, this thesis does not fabricate recognition accuracy, usability scores, learning gains or end-to-end physical latency. Instead, it reports auditable artifact-level results and provides a complete measurement protocol for technical and educational validation.

Keywords: sign-language learning, haptic feedback, wearable IoT, ESP32, MediaPipe, hand landmarks, interactive practice, Web Bluetooth, computer vision, O/L ICT

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List of Abbreviations

Abbreviation	Meaning
ASL	American Sign Language
BLE	Bluetooth Low Energy
API	Application Programming Interface
ESP32	Espressif ESP32 microcontroller family
ERM	Eccentric Rotating Mass
GATT	Generic Attribute Profile
ICT	Information and Communication Technology
IoT	Internet of Things
LRA	Linear Resonant Actuator
OLED	Organic Light-Emitting Diode
SUS	System Usability Scale
TP/FN/FP/TN	True Positive / False Negative / False Positive / True Negative

1.0 Introduction

1.1 Background

Digital learning environments can improve access to instructional content for learners who experience barriers with conventional audio-centred teaching. For Deaf and hard-of-hearing learners, visual communication is central, but interactive practice still faces a feedback problem such as a learner can make an incorrect hand configuration, repeat the same error and receive delayed or purely visual correction. Recent reviews of assistive and immersive technologies report growing use of gesture recognition, interactive sign teaching, visual cues and personalized learning, but also highlight hardware limitations, inconsistent recognition performance and the need for more inclusive evaluation [1]-[4].

Haptic feedback offers a complementary channel. Instead of requiring the learner to look continuously at a screen, a wearable device can communicate a discrete event through vibration. A systematic review of vibrotactile feedback in motor learning found that real-time vibrotactile cues are commonly used as a multimodal supplement and can support learning under suitable feedback designs [5]. Sign-language learning systems have also explored wearable gloves and vibration motors, showing the feasibility of combining gesture recognition with tactile cues [6]. These findings motivate the exploration of a simpler, lower-cost wristband rather than a sensor-rich glove.

My component of the SignLearn AI project addresses this opportunity by combining an interactive sign-practice arena with an ESP32-based wristband. The arena presents ICT-related practice terms, observes the learner through a webcam, estimates hand landmarks using MediaPipe, checks predefined hand-shape requirements, records pass and error events, and triggers haptic feedback through the wristband pathway. The system also includes OLED messages, notification presets, event history, progress tracking and a virtual wristband for development and demonstration.

1.2 Individual Component

This individual report focuses on the Smart Haptic Wristband IoT and Sign Practice Arena. The contribution analyzed and documented here comprises the practice workflow, hand-landmark evaluation logic, wristband notification model, ESP32 prototype firmware, OLED and haptic output, backend event and progress handling, browser communication path, engineering measurements, and the comparative evaluation methodology. Dependencies on the other platform components are treated as integration interfaces rather than claimed as independent work.

1.3 Problem Statement

The core problem is the lack of immediate, multimodal feedback during sign-language practice for an educational learner using a conventional webcam. Visual feedback alone can be overlooked or can require repeated attention shifts. At the same time, many sophisticated recognition systems rely on larger datasets, machine-learning classifiers, or specialized wearable gloves that increase development and hardware complexity [6]-[8]. There is therefore a practical engineering question: can a low-cost webcam-based practice environment be combined with a simple wearable haptic device to provide timely corrective feedback without requiring a complex full-body recognition system?

Within the supplied implementation, this question is addressed through a rule-based prototype rather than a learned end-to-end sign-language recognizer. This makes the component suitable for a final-year engineering project, but it also means that the research must evaluate the prototype honestly and distinguish implementation evidence from unmeasured performance.

1.4 Research Gap

Recent literature shows three partially disconnected streams. First, computer-vision systems can provide real-time hand tracking and sign recognition using hand landmarks and lightweight classifiers [7]-[10]. Second, educational interfaces increasingly use interactive practice, avatars, and multimodal feedback [2]-[4]. Third, haptic learning research demonstrates that vibration can support motor learning and real-time feedback [5]. However, recent reviews of sign-language education note that relatively few systems systematically integrate haptic feedback with a visual practice environment, and they call for better evaluation of accessibility, hardware choices, and learning outcomes [3].

The practical gap addressed by Component 05 is therefore the integration layer: a low-cost wrist-worn actuator is connected to a browser-based practice workflow so that a sign attempt produces an immediate physical cue. The research contribution is framed as an integrated visual-haptic prototype plus an empirical evaluation strategy. It is not claimed to be a novel state-of-the-art sign classifier.

1.5 Research Objectives

1. Design and implement an interactive sign-practice arena for a defined set of ICT-related sign keywords.
2. Implement a lightweight hand-analysis pipeline using MediaPipe hand landmarks and deterministic hand-shape rules.
3. Design and implement an ESP32-based wristband with OLED and vibration output for learner feedback.
4. Integrate the practice arena, backend progress services, notification history, and wearable feedback paths.
5. Evaluate the functional reliability and timing of gesture evaluation and haptic feedback under controlled trials.
6. Investigate whether adding haptic feedback to visual feedback improves practice performance, feedback awareness, and perceived usability compared with visual-only practice.

1.6 Research Questions

1. Q1: How effectively can the implemented MediaPipe-and-rule-based pipeline detect the predefined hand-shape configurations used by the Component 05 practice curriculum?
2. Q2: How reliably and quickly can a pass/error event produce a corresponding wristband feedback action?
3. Q3: Does the addition of haptic feedback improve practice outcomes such as pass rate, number of retries, and completion time compared with visual-only practice?
4. Q4: How do learners perceive the usefulness, immediacy, comfort, and usability of the visual-haptic system?
5. RQ5: What engineering trade-offs arise from the current prototype communication architecture, particularly the separation between browser Web Bluetooth code and the Wi-Fi polling path used by the supplied firmware?

1.7 Scope

- The scope is limited to the supplied Component 05 prototype and its defined eight ICT keywords: computer, hardware, software, database, network, internet, email, and security.
- The current recognition approach is limited to predefined hand configurations based primarily on finger-extension counts from 21 MediaPipe landmarks.
- The physical prototype consists of an ESP32, 128x64 SSD1306 OLED, vibration motor, and push button as represented in the supplied firmware.
- The thesis evaluates the system as a learning-support prototype; it does not claim clinical, diagnostic, or universally valid sign-language recognition.
- Any study involving minors or school learners requires the applicable institutional approvals and consent procedures. For a rapid pilot, adult volunteers are recommended.

1.8 Engineering and Research Contributions

- A low-cost architecture that adds haptic feedback to a webcam-based sign-practice interface.
- A deterministic landmark-based evaluator suitable for constrained ICT sign exercises without requiring a training dataset.
- An integrated backend model for wristband configuration, notification events, learning progress, and completion tracking.
- A practical evaluation protocol covering recognition, communication reliability, latency, usability, and comparative learning performance.
- An explicit implementation audit that identifies limitations and separates intended architecture from the currently supplied firmware behavior.

1.9 Thesis Organization

Chapter 2 reviews relevant literature. Chapter 3 defines the research design and evaluation methodology. Chapter 4 analyzes and designs the system. Chapter 5 documents the implementation and algorithms. Chapter 6 provides the testing strategy and the evidence currently available from the supplied project, together with tables to be populated with actual measurements. Chapter 7 interprets the findings and limitations, while Chapter 8 concludes the study and proposes future work.

1.10 Evidence Boundary of the Study

Three evidence categories are distinguished throughout this thesis. Implementation facts are verified directly from the supplied repository. Derived engineering quantities are calculated from fixed source-code parameters. Empirical outcomes - including participant accuracy, latency, learning improvement, comfort and usability scores - require runtime or human-subject measurements and are not inferred from interface defaults. This distinction is used consistently so that the report remains auditable and defensible.

2.0 Literature Review

2.1 Technology-Assisted Sign-Language Education

Technology-assisted education for Deaf and hard-of-hearing learners spans video, animation, computer vision, augmented/virtual reality, captioning, and interactive practice. A recent systematic review of assistive technologies for Deaf communication identified gesture recognition and sign teaching as recurring application areas [2]. A 2026 systematic literature review of immersive educational technologies similarly reports benefits from multimodal instructional access while noting hardware, accessibility, and implementation constraints [4]. These reviews suggest that technology can broaden access, but effectiveness depends on interaction design and evidence from actual users.

2.2 Computer Vision and Hand Landmark Tracking

MediaPipe Hands demonstrated accurate, real-time hand tracking using a palm detector followed by a hand-landmark model, enabling 21 hand landmarks and efficient on-device processing [7]. The approach is attractive for educational prototypes because it provides structured landmarks without requiring a custom hand-detection model. Follow-on work has used MediaPipe landmarks as inputs to segmentation and sign-recognition pipelines [8]-[10].

The main advantage for Component 05 is computational simplicity. Instead of training an end-to-end classifier, the system derives a small set of geometric states from the landmarks. This is explainable and easy to debug. The trade-off is reduced robustness to hand orientation, occlusion, camera viewpoint, temporal movement, and sign variations.

2.3 Sign Recognition Approaches

Research systems range from rule-based geometric methods to machine-learning classifiers and sequence models. CopyCat, for example, explored sign-language recognition to support children learning language, using MediaPipe poses and temporal modeling [11]. Other recent systems combine MediaPipe skeletons with point-cloud networks or Random Forest classifiers and report high recognition performance under their own controlled datasets [9], [10]. Such results should not be compared directly with Component 05 because the evaluation conditions, sign vocabulary, participants, data collection, and model complexity differ substantially.

This Component evaluator is intentionally lightweight. It does not contain a learned classifier. It sorts detected hands horizontally, counts extended fingers using landmark heuristics, and checks whether the observed counts equal project-defined requirements. This makes the system a constrained practice evaluator rather than a general ASL recognition engine.

2.4 Haptic and Vibrotactile Feedback

Vibrotactile feedback is established as a multimodal feedback channel in motor learning. A systematic review of 24 peer-reviewed studies found that real-time vibrotactile feedback is frequently combined with visual information and can support motor learning when the feedback design is appropriate [5]. The review also highlights the importance of actuator location, timing, and the relationship between feedback and the learner's action.

For my component , the wristband maps two high-level events to vibration patterns. A correct sign produces a short confirmation pulse, while an incorrect sign produces a repeated pulse intended to attract attention and encourage correction. The mapping is deliberately discrete rather than continuous because the prototype does not provide a calibrated physical vibration-intensity measurement.

2.5 Wearable Devices for Sign-Language Learning

Wearable sign-language learning systems have also used gloves with bend/gesture sensors and vibration motors. Kim and Baek developed a wearable-glove learning system using neural-network-based recognition and haptic cues [6]. Such systems can achieve direct hand sensing, but they introduce additional sensors, calibration requirements, and cost. The wristband approach in my component moves the sensing burden back to the camera while retaining tactile feedback as the learning channel.

2.6 Interactive Practice, Avatars and Feedback Timing

Interactive learning benefits from repeated low-cost practice, immediate feedback, and understandable correction. The SignLearn AI practice arena follows this pattern by placing an avatar demonstration beside the camera evaluator. A learner chooses a keyword, observes the target hand configuration, performs it, maintains the configuration briefly, and receives success or correction feedback. This structure creates a closed learning loop even though the current recognition model is deliberately simple.

2.7 Web Bluetooth and Browser Constraints

The supplied frontend includes a Web Bluetooth controller based on GATT services and characteristics. Web Bluetooth provides browser access to Bluetooth Low Energy GATT peripherals, but it remains a constrained web platform capability with availability and permission limitations across browsers and operating systems [12]. This matters for the research because the browser BLE path should be tested separately from the Wi-Fi firmware path, and the thesis should not treat the presence of BLE JavaScript as proof that the current ESP32 firmware implements a compatible GATT service.

2.8 Research Gap and Conceptual Framework

The literature supports the conceptual chain used in this thesis: a sign-learning action is observed through computer vision; the action is mapped to a constrained gesture state; the state produces immediate multimodal feedback; the learner uses that feedback to adjust the next attempt. The missing empirical link is whether the added tactile channel measurably improves practice outcomes under controlled conditions. This becomes the central experimental proposition of the thesis.

3.0 Research Methodology

3.1 Research Paradigm

The study follows a Design Science Research orientation. Design science treats an information-system artifact as both a practical solution and an object for rigorous evaluation [13]. This is suitable because my component produces a tangible artifact (the wristband and integrated software), tests it in controlled conditions, and derives lessons about usability, reliability, and educational feedback.

3.2 Research Phases

Phase	Activity	Output
1. Problem definition	Identify feedback limitations in visual-only sign practice	Problem statement + RQs
2. Literature grounding	Review sign education, hand tracking, haptics, wearable learning	Research gap + hypotheses
3. Artifact design	Define hardware, software, interaction and data architecture	System design
4. Implementation	Build practice arena, evaluator, backend, wristband prototype	Working prototype
5. Verification	Run functional, communication and recognition tests	Technical evidence
6. User evaluation	Compare visual-only and visual-haptic practice	Performance + usability data
7. Analysis	Compute metrics and interpret observations	Results + discussion
7. Analysis	Compute metrics and interpret observations	Results + discussion

3.3 Research Design

A small within-subject pilot is recommended because it reduces the variability between participants and is practical within a final-year project. The same participant completes matched sign-practice tasks under two conditions: Condition A uses visual feedback only, while Condition B adds wristband haptic feedback. The order should be counterbalanced (half visual-first, half haptic-first) where the sample size permits, reducing simple order effects.

Recommended pilot sample: 20 adult volunteers, with a smaller sample reported explicitly as a pilot if time or recruitment prevents reaching the target. This is an engineering usability study, not a population-level claim about Deaf education. Any study involving minors or school learners requires appropriate ethics review and informed consent/assent procedures.

3.4 Variables

Independent	Feedback condition	Visual only vs visual + haptic
Dependent	Recognition outcome	Correct/incorrect classification against a predefined expected hand state
Dependent	Retries	Number of attempts before pass
Dependent	Completion time	Seconds from attempt start to successful completion
Dependent	Feedback latency	Time from evaluation event to feedback activation/display
Dependent	Usability	SUS score plus custom haptic-awareness items
Controlled	Camera, lighting, distance	Keep approximately constant within the test session

3.5 Technical Evaluation Metrics

Recognition metrics should be calculated from trial-level outcomes, not from the fixed "accuracy" values currently displayed by the interface.

Metric	Formula	Interpretation
Accuracy	(TP + TN) / (TP + TN + FP + FN)	Overall classification correctness
Precision	TP / (TP + FP)	How many detected positives were correct
Recall	TP / (TP + FN)	How many expected positives were detected
F1	2 * Precision * Recall / (Precision + Recall)	Balance of precision and recall
False rejection rate	FN / (TP + FN)	Correct gestures rejected
False acceptance rate	FP / (FP + TN)	Incorrect gestures accepted
Event reliability	successful feedback events / requested feedback events	Wearable actuation reliability
Latency	t_feedback - t_trigger	End-to-end feedback delay

3.6 Learning-Performance Analysis

For each participant, calculate mean or median retries and completion time separately for each feedback condition. If data are approximately normally distributed, a paired t-test may be used for the two-condition comparison. If the sample is small or non-normal, the Wilcoxon signed-rank test is a safer non-parametric alternative. Report an effect size where possible. Because this is a pilot, confidence intervals and descriptive statistics are more informative than relying only on p-values.

3.7 Usability Evaluation

The System Usability Scale (SUS) provides a short, standardized usability measure [14]. The conventional SUS score ranges from 0 to 100 after reversing the negatively worded items and applying the standard 2.5 multiplier. The questionnaire in Appendix C adds project-specific items for vibration noticeability, comfort, perceived immediacy, and usefulness. The custom questions should be reported separately rather than incorrectly merged into the SUS score.

3.8 Experimental Procedure

1. Explain the task and obtain consent.

2. Calibrate camera position and lighting; record participant code only.

3. Provide a short familiarization phase with two practice keywords.

4. Run Condition A on a fixed subset of keywords; record attempts, pass outcome, completion time and feedback latency.

5. Provide a short break or neutral task.

6. Run Condition B using the same task structure and counterbalanced order.

7. Collect SUS and custom feedback ratings.

8. Export or manually record technical event logs, then compute metrics using the formulas above.

9. Preserve raw data separately from processed tables so that the reported results are auditable.

3.9 Threats to Validity

- Internal validity: participants may improve simply through repetition; counterbalancing and matched trials reduce this effect.
- Construct validity: the current rule-based evaluator measures predefined hand configurations, not full semantic ASL competence.
- External validity: a small adult pilot does not establish effectiveness for school-age Deaf learners.
- Technical validity: webcam lighting, hand occlusion, skin-background contrast, and camera placement can affect landmarks.
- Integration validity: the current firmware and browser BLE controller use different communication paths; these must be tested and reported separately.
- Learning effect: novelty of the wristband may temporarily increase engagement; longer-duration studies are required to establish retention.

3.10 Ethical Considerations

Participants should receive an information sheet explaining what is recorded, why it is recorded, how long it is stored, and how the data are anonymized. Webcam images should not be retained unless specifically approved. Participant identifiers should be replaced with codes. Any study involving minors, Deaf school learners, or identifiable video data must follow the university's applicable ethics and consent process.

3.11 Measurement Protocol and Data Quality

Every recognition event should have an independent expected label and a unique trial identifier. This allows true positives, false positives, false negatives and true negatives to be counted without trusting client-supplied confidence. For haptic testing, record trigger time, backend acknowledgement time and physical observation time separately. For participant data, use pseudonymous codes and do not retain webcam images unless specifically approved by the institution.

For a practical engineering validation, a minimum of 160 recognition trials is recommended: 8 keywords x 10 correct presentations + 10 deliberately incorrect presentations. For feedback reliability, at least 30 success events and 30 error events should be observed. These are project-level sampling recommendations, not universal statistical standards.

3.12 General Engineering Measurement Conditions

To make my component measurements reproducible, the proposed validation environment should be controlled even when the target metric is simple. Recognition tests should use a fixed webcam position and field of view, a stable background, consistent participant distance, and representative indoor lighting. Timing should be recorded from the same clock source wherever possible. A recognition event is defined from the instant a frame is evaluated to the instant the corresponding user-visible or haptic response is observed; software processing time and end-to-end user-perceived latency should therefore be reported separately. These conditions are measurement controls, not claims that the current prototype has already achieved them.

Measurement condition	Recommended controlled value	Reason
Camera resolution	640 × 480 pixels	Matches the current webcam constraint used by the Practice Arena.
Recognition sampling	80 ms minimum detection interval	Matches the current front-end throttle; corresponds to a nominal 12.5 Hz schedule.
UI state update	100 ms minimum interval	Matches the current UI synchronization loop; corresponds to a nominal 10 Hz schedule.
Hold-to-confirm	1.2 s	Matches the implemented pass confirmation window and reduces accidental single-frame passes.
Pass cooldown	2.0 s	Matches the implemented cooldown between successful evaluations.
Firmware configuration polling	6 s	Matches the ESP32 firmware polling interval for wristband configuration.
Firmware status polling	15 s	Matches the ESP32 firmware status reporting interval.
Recognition sample set	160 recommended trials	8 current keywords × 10 correct + 10 deliberately incorrect trials.

4.0 System Analysis and Design

4.1 System Context

My component is a subsystem inside SignLearn AI. Its purpose is to connect a learner-facing interactive practice arena to a physical wearable feedback channel. The project repository separates frontend, backend, hardware firmware, and component documentation. The supplied implementation uses React/Vite/Tailwind on the frontend, FastAPI with Motor/MongoDB on the backend, MediaPipe Tasks Vision for hand landmarks, and ESP32 firmware for the wristband.

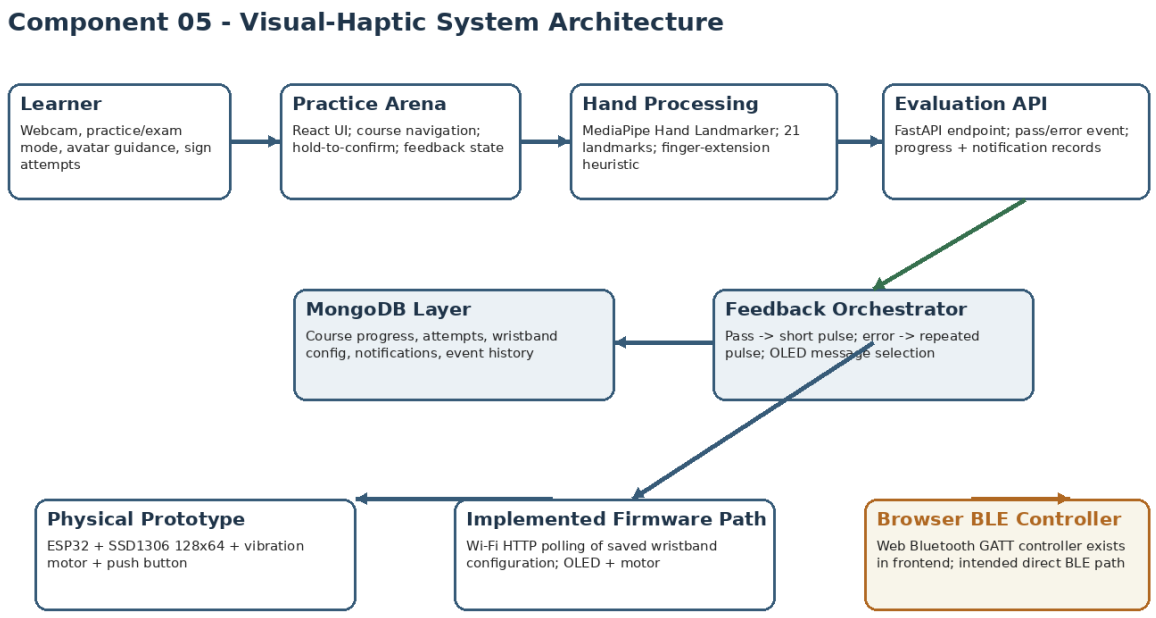


Figure 4.1. Architecture

4.2 Architecture Consistency and Integration Risk

The repository contains two communication concepts: a browser-side Web Bluetooth GATT controller and an ESP32 firmware implementation based on Wi-Fi HTTP polling. The firmware polls the configuration endpoint every 6 seconds and reports status every 15 seconds; it does not implement the Nordic UART-style GATT service expected by the browser controller. Accordingly, these should be treated as alternative prototype routes until a single transport is implemented and verified end-to-end.

4.3 Functional Requirements

ID	Requirement	Verification
FR-01	Display a selectable set of sign-practice keywords.	UI inspection + course module API
FR-02	Capture webcam frames and detect up to two hands.	Runtime test
FR-03	Evaluate predefined hand-count requirements.	Recognition test
FR-04	Require a brief stable hold before pass.	Hold-timer test
FR-05	Persist pass/error events and progress.	API/database test
FR-06	Generate distinct success and error haptic patterns.	Wristband functional test
FR-07	Display short feedback messages on OLED.	Hardware test
FR-08	Maintain wristband configuration and event history.	Backend CRUD/history test
FR-09	Provide a development simulation path when hardware is unavailable.	Virtual wristband test
FR-10	Support a browser BLE controller for a compatible GATT device.	Separate BLE compatibility test

4.4 Course Model

The backend course service currently defines two modules and eight keywords. Both modules use a passing threshold of 70 in the course service. Module 1 contains computer, hardware, software, and database. Module 2 contains network, internet, email, and security. The service considers the course complete when all eight keywords have been completed. The repository also contains a model default with totalKeywords=15 and passingThreshold=75, which is inconsistent with the service implementation and should be corrected before a final release.

4.5 Gesture Representation

Each keyword is represented by a project-defined left/right hand finger-count specification. Examples include both hands with five extended fingers for “computer,” left hand five and right hand two for “software,” and both hands with zero extended fingers for “security.” The evaluator mirrors coordinate, sorts hands horizontally, counts extended fingers, and checks whether the observed count equals the required count.

Keyword	Left requirement	Right requirement	Evaluation rule
computer	5	5	Both hands must match
hardware	5	0	Left hand match; right hand absent/closed
software	5	2	Both hand counts must match
database	3	3	Both hand counts must match
network	3	3	Both hand counts must match
internet	5	5	Both hands must match
email	3	2	Both hand counts must match
security	0	0	Both hands closed

5.0 Implementation

5.1 Frontend Practice Arena

The Practice Arena is implemented using React components. The main practice screen combines an avatar demonstration area with a camera evaluator. Practice and exam modes are supported. The exam mode can hide the demonstration avatar to reduce visual assistance. The interface also presents hand-shape and movement guidance text.

5.2 MediaPipe Hand Landmark Pipeline

CameraSignEvaluator initializes MediaPipe Tasks Vision HandLandmarker in VIDEO mode with numHands=2 and a GPU delegate. Webcam input is configured at approximately 640x480. Detection is throttled to roughly one call every 80 ms, corresponding to about 12.5 landmark evaluations per second under the implementation logic, while the browser animation frame continues to update UI state.

5.3 Finger-Extension Heuristic

For non-thumb fingers, the implementation compares the fingertip y-coordinate against the proximal-interphalangeal landmark y-coordinate after the display coordinate transformation; the thumb uses a distance ratio against the wrist and thumb metacarpophalangeal landmark. The result is a binary extended/not-extended state for each finger. This is computationally inexpensive and interpretable, but it is sensitive to viewpoint and does not model sign trajectories or joint angles in a general way.

5.4 Hand Sorting and Matching

Detected hands are sorted horizontally into screen-left and screen-right positions. The required finger counts are then matched to the corresponding hand. A keyword is accepted only when the detected configuration matches the required specification. The matching mechanism is deterministic, which supports repeatable unit tests but limits vocabulary generalization.

5.5 Hold-to-Confirm Mechanism

When the observed configuration matches the requirement, the interface starts a stable-hold timer. The implementation uses approximately 1.2 seconds as the hold duration. Progress is displayed during the hold. This design attempts to reduce accidental single-frame acceptance and encourages the learner to maintain the target hand configuration briefly.

5.6 Backend Evaluation and Progress

The frontend sends a course evaluation event to the backend. The current service can accept client-provided isCorrect and confidence fields and uses them directly to determine the pass outcome. On a pass it creates a “Sign Success Alert,” stores “SIGN PASSED,” requests the Short Pulse pattern, updates progress, and returns the next keyword. On an error it creates a “Wrong Sign Alert,” stores “RETRY SIGN,” requests Repeated Pulse, increments mistake count and wristband trigger count, and returns the next keyword.

5.7 Wristband Firmware

The supplied firmware targets an ESP32 and uses Wi-Fi, HTTPClient, ArduinoJson, Wire, and Adafruit SSD1306. It drives the motor through GPIO 16, reads a push button on GPIO 4, and displays status and feedback on a 128x64 OLED. The firmware polls the wristband configuration endpoint every six seconds and sends a status/notification event every fifteen seconds. It implements six timed vibration patterns through digital motor on/off control.

The six firmware pattern timings are approximately: Short Pulse 200 ms; Double Pulse 200 ms + 150 ms gap + 200 ms; Long Pulse 1000 ms; Short+Long 200 ms + 150 ms gap + 800 ms; Repeated Pulse five cycles of 200 ms with 130 ms gaps; Emergency Pulse eight cycles of 120 ms with 80 ms gaps. These timings should be treated as prototype pattern definitions rather than calibrated perceptual intensities.

Haptic Alert Mapping Implemented in the Backend

Event	OLED message	Pattern	Intensity field	Purpose
Sign Success Alert	SIGN PASSED	Short Pulse	50	Immediate positive confirmation
Wrong Sign Alert	RETRY SIGN	Repeated Pulse	85	Immediate correction request

Figure 5.1. Backend haptic alert mapping and its current physical interpretation.

5.8 Browser Web Bluetooth Controller

The frontend contains a Web Bluetooth controller using a Nordic UART Service-style UUID family. It requests a compatible GATT device, connects to the primary service, obtains TX/RX characteristics, and writes JSON vibration commands. The controller includes action, pattern, OLED text, intensity, and duration fields.

5.9 Communication Architecture Mismatch Identified by Audit

The supplied browser BLE controller and ESP32 firmware do not currently implement the same transport. The firmware imports Wi-Fi/HTTP libraries and contains no BLE GATT service implementation, while the browser controller expects a BLE GATT server. Therefore, the safest technical description is that the project contains two communication strategies: (1) the currently supplied firmware path, which uses Wi-Fi HTTP polling of backend configuration; and (2) a frontend Web Bluetooth path intended for a compatible BLE firmware implementation. This finding should be acknowledged rather than hidden.

5.10 Virtual Wristband and Development Paths

The frontend can emulate wristband feedback, which is useful when physical hardware is unavailable. The evaluator also contains development/test controls that can force success or error paths. These are appropriate for integration testing but must be disabled or excluded from participant experiments because they bypass the recognition pipeline.

5.11 Implementation Audit Summary

Audit item	Observed implementation	Research implication
Course vocabulary	8 keywords in 2 modules	Use 8 as the actual current scope
UI “accuracy”	Fixed values such as 97, 40, 20, 98	Cannot be reported as measured recognition accuracy
Backend trust	Client pass/confidence accepted	Independent ground-truth evaluation required
Firmware transport	Wi-Fi HTTP polling	Do not claim current firmware is BLE-compatible
Browser transport	Web Bluetooth GATT controller	Needs compatible BLE firmware for direct use
Motor intensity	Application field exists; firmware uses digital ON/OFF	Intensity is configuration metadata, not calibrated amplitude
Device battery/status	Service initializes fixed prototype values	Do not report as measured battery telemetry

Audit item	Observed implementation	Research implication
Sign definitions	Project code labels them as ASL	Linguistic validity should be reviewed by a sign-language expert
Metadata mismatch	Model default says 15 keywords; service uses 8	Correct before final release

5.12 Verified Engineering Parameter Sheet

The following values are observable or directly derivable from the supplied source archive. They are implementation measurements, not educational outcome claims.

Parameter	Value	Evidence	Meaning
Component 05 source files	24 files; approx. 5,559 lines	Repository count	Implementation scope
Course scope	2 modules; 8 keywords	Backend service	Current curriculum size
Maximum tracked hands	2	MediaPipe configuration	Bilateral practice support
Landmarks per hand	21	MediaPipe hand model	Basis for finger heuristics
Detection throttle	80 ms	CameraSignEvaluator	Nominal 12.5 detection cycles/s
UI state sync	100 ms	CameraSignEvaluator	Nominal 10 state updates/s
Hold-to-confirm	1.2 s	CameraSignEvaluator	Stability interval before pass
Pass cooldown	2.0 s	CameraSignEvaluator	Reduces repeated pass events
ESP32 configuration	6.0 s	SmartHapticWristband.ino	Potential polling

Parameter	Value	Evidence	Meaning
polling			delay in Wi-Fi route
ESP32 status interval	15.0 s	SmartHapticWristband.ino	Status cadence
OLED	128 x 64 pixels	Firmware	Learner-facing text output
Motor output	GPIO16 digital ON/OFF	Firmware	No calibrated analogue intensity
Button input	GPIO4 / INPUT_PULLUP	Firmware	Local test mode
Alert presets	9	wristband_alert_presets.json	Reusable event mapping
Named vibration patterns	6 + fallback	Firmware	Discrete timing patterns

Derived vibration durations are: Short Pulse 200 ms; Double Pulse 550 ms; Long Pulse 1,000 ms; Short + Long 1,150 ms; Repeated Pulse approximately 1,520 ms; and Emergency Pulse approximately 1,520 ms. The separate API duration field must therefore not be interpreted automatically as the actual physical motor-on time.

6.0 Testing, Evaluation and Results

6.1 Evidence Classification

The results are separated into (a) source-verified implementation results, (b) derived engineering measurements, and (c) empirical user or hardware measurements that require controlled execution. The supplied project archive contains evidence for the first two categories but does not contain a completed participant dataset or instrumented physical latency log.

6.2 Artifact-Level Verification Results

Verification item	Observed evidence	Status
Course model	Two modules containing eight keywords are defined in the backend service.	Verified
Hand landmark setup	HandLandmarker configured for VIDEO mode with numHands = 2.	Verified
Hold-to-confirm	Frontend uses approximately 1.2 seconds of continuous match.	Verified
Success feedback	Pass path requests Short Pulse / SIGN PASSED.	Verified
Error feedback	Failure path requests Repeated Pulse / RETRY SIGN.	Verified
OLED display	SSD1306 128x64 is initialised in firmware.	Verified
Motor output	GPIO16 is controlled with digital HIGH/LOW.	Verified
Firmware transport	Wi-Fi HTTP polling of wristband configuration.	Verified

Verification item	Observed evidence	Status
Browser BLE path	Web Bluetooth GATT controller exists in frontend.	Verified
End-to-end BLE compatibility	No corresponding BLE GATT service implementation found in supplied firmware.	Not verified
UI accuracy values	97/98/40/20 are state values in frontend logic, not measured accuracy.	Verified as code behaviour; not an empirical metric

6.3 Derived Timing Measurements

Metric	Calculation	Derived value	Interpretation
Nominal detection cadence	1 / 0.080 s	12.5 Hz	Scheduled landmark evaluation rate
UI state cadence	1 / 0.100 s	10 Hz	Scheduled matching/state updates
Hold interval	1.2 s	1.2 s	Required continuous match before pass
Firmware configuration poll	1 / 6 s	0.167 Hz	Polling frequency of physical Wi-Fi route
Firmware status report	1 / 15 s	0.067 Hz	Status event frequency
Short Pulse duration	200 ms	0.20 s	Single confirmation pulse
Double Pulse duration	200+150+200 ms	0.55 s	Two confirmation pulses
Repeated Pulse duration	5x200 + 4x130 ms	1.52 s	Correction pattern

6.4 Software Verification Results

Static verification of the supplied non-JSX source files was performed during thesis preparation. The Component 05 Python modules passed Python compilation checks; the wristband alert preset JSON parsed successfully; and the standalone JavaScript service/utility files passed Node syntax checks. These checks establish syntax-level integrity only and do not substitute for browser runtime, hardware or participant testing.

Verification	Result	Meaning
Python component files	PASS	No Python syntax errors detected in the inspected Component 05 modules
Wristband preset JSON	PASS	Preset file is valid JSON
Standalone JavaScript files	PASS	No syntax errors detected in inspected non-JSX service/utility files
JSX runtime behaviour	Not measured	Requires browser build and execution
ESP32 firmware compile/flash	Not measured	Requires Arduino/ESP32 toolchain and physical board

6.5 What Is Not Measured in the Supplied Evidence

- Recognition accuracy, precision, recall and F1 cannot be reported as empirical results without a labelled trial dataset.
- End-to-end haptic latency cannot be reported without timestamp instrumentation and physical observation.
- Physical vibration intensity and comfort threshold are not calibrated; the intensity field is configuration metadata while the motor is digitally switched.
- Battery endurance and true connectivity status are not measured telemetry in the supplied prototype.
- SUS and learner preference scores are unavailable because no completed questionnaire dataset was included.

- Educational effectiveness for Deaf or school-age learners cannot be concluded from source code alone.

6.6 Recommended Engineering Test Matrix

Test class	Recommended minimum	Measures	Project target
Recognition positives + negatives	160 trials	Accuracy, precision, recall, F1, FAR, FRR	>=90% repeatability under controlled setup
Success feedback	30 events	Success count, misses, latency	>=95% successful events
Error feedback	30 events	Success count, misses, latency	>=95% successful events
Progress persistence	10 reload cycles	State retained after refresh	100% preservation
Course completion	3 clean runs	All eight keywords and completion state	3/3 successful runs
Browser BLE route	10 connect/write cycles	Connect, write, disconnect recovery	Report separately from Wi-Fi route

The targets above are project-level acceptance targets, not universal standards. A failed target remains a useful research result and should be discussed rather than excluded.

6.7 Comparative Learning Evaluation

The intended research comparison is a within-subject design. Each participant performs matched practice tasks in two conditions: visual feedback only, and visual feedback plus wristband haptics. The primary outcomes are retries per sign, completion time, pass rate and feedback-awareness rating. When paired differences are approximately normal, a paired t-test can be used; otherwise the Wilcoxon signed-rank test is appropriate. Median and interquartile range should be reported for skewed measures, together with an effect size and confidence interval where feasible.

Outcome	Visual-only	Visual + haptic	Analysis
Retries/sign	Measure during study	Measure during study	Paired comparison
Completion time	Measure during study	Measure during study	Paired comparison
Pass rate	Measure during study	Measure during study	Descriptive + paired comparison
Feedback awareness	Likert rating	Likert rating	Condition comparison
SUS	Post-use measure	Post-use measure	Descriptive usability score

6.8 Interpretation of the Current Result Set

The current evidence demonstrates implementation feasibility and evaluation readiness. It does not demonstrate that the wristband improves learning. The most defensible research claim is that Component 05 provides a constrained, explainable visual-haptic interaction that can now be evaluated systematically. Any numerical statement about recognition, latency, usability or learning gain must come from actual measurements collected under the protocol above.

6.9 General Measurement and Acceptance Guide

Because the supplied archive does not contain a completed participant or physical-hardware result set, this thesis distinguishes measured values from engineering acceptance targets. The following guide should be used when the final validation run is performed. A target is not a result: it is a predefined criterion against which an observed result can be judged. This prevents favourable numbers from being selected after the experiment.

Metric	Operational measurement	Project-level reference / target	Reporting rule
Recognition accuracy	(TP + TN) / all labelled trials	≥ 90% repeatability under controlled conditions	Report the measured value with the trial count; do not replace it with UI confidence values.
Precision	TP / (TP + FP)	≥ 90% suggested engineering target	Report TP, FP and the denominator.
Recall	TP / (TP + FN)	≥ 90% suggested engineering target	Report TP, FN and the denominator.
F1 score	2PR / (P + R)	≥ 0.90 suggested engineering target	Report together with precision and recall.
Haptic event reliability	successful physical activations / triggered events	≥ 95% suggested engineering target	Separate communication success from physical vibration success.
End-to-end latency	physical observation time − trigger time	Measure and report median, IQR and maximum	Do not infer this from the 80 ms UI throttle or 6 s polling alone.
Course progress persistence	correctly restored records / reload trials	100% for the tested persistence scenarios	Report the number of successful reloads and failures.
SUS usability	standard SUS scoring procedure	Descriptive comparison; no invented threshold	Report score, sample size and dispersion.
Learning effect	paired change in retries/time/pass rate	Hypothesis test rather than arbitrary pass/fail target	Use paired t-test only when assumptions are acceptable; otherwise use Wilcoxon signed-rank.

7.0 Discussion

7.1 Interpretation of the Artifact

The main engineering achievement of Component 05 is integration. The learner does not need a separate app for practice and feedback: the practice arena, course state, backend event handling, virtual simulation, and physical prototype are designed as one workflow. This aligns with the literature's broader trend toward multimodal and interactive support for Deaf and hard-of-hearing learners [2]-[5].

At the same time, the prototype deliberately prioritizes transparency and low implementation complexity over recognition sophistication. Landmark-based finger counts are easy to explain and test, but they do not capture full sign semantics, hand orientation in all poses, motion trajectories, co-articulation, facial expression, or signer variation. Therefore the most defensible claim is that the system recognizes constrained practice configurations, not that it solves general ASL recognition.

7.2 Interpretation of Haptic Feedback

The haptic design follows a simple binary interaction principle: success is mapped to a short positive pulse, whereas an error produces a more salient repeated pulse. This is consistent with the role of vibrotactile feedback as an additional channel rather than a replacement for visual feedback [5]. The main empirical question is not whether vibration can exist technically, but whether its timing and salience improve the learner's response without creating distraction or fatigue.

7.3 Comparison with Related Work

Compared with high-accuracy research recognizers [9], [10], the Component 05 model is much simpler and narrower. Compared with glove-based systems [6], it requires fewer body-worn sensors. Compared with purely visual practice environments, it adds a physical feedback channel. Therefore its contribution is primarily integration, accessibility-oriented interaction, and an implementation pathway that can be measured and improved.

7.4 Key Limitations

- The current hand evaluator is heuristic and vocabulary-constrained.
- Hard-coded UI accuracy values are not empirical confidence estimates.
- The backend currently trusts client-supplied correctness and confidence fields.

- The physical firmware supplied in the archive uses Wi-Fi polling rather than the browser's BLE GATT path.
- Vibration intensity is represented in configuration fields but is not physically calibrated in firmware.
- Device battery and connection status shown by the service are prototype values rather than sensor telemetry.
- The project's sign labels should be linguistically validated by a qualified sign-language expert.
- No completed participant study is present in the archive, so educational effectiveness cannot be claimed from the current source alone.

7.5 Future Improvements

- Replace finger-count heuristics with a validated landmark classifier that uses joint angles, distances, handedness and, where appropriate, temporal features.
- Implement a single consistent communication architecture: either BLE GATT in firmware or an explicit Wi-Fi/WebSocket event channel.
- Add real event queues rather than polling only the saved configuration record.
- Implement PWM-based motor control with experimentally calibrated intensity levels and participant comfort thresholds.
- Add objective timestamp instrumentation for end-to-end latency.
- Expand vocabulary and include temporal signs only after collecting representative data.
- Conduct a longitudinal study with target learners after the prototype passes technical validation.
- Add accessibility review with Deaf/sign-language users and expert teachers.

8.0 Conclusion and Future Work

8.1 Conclusion

My component demonstrates a practical architecture for combining computer-vision-based sign practice with wearable haptic feedback. The supplied implementation includes an interactive React practice arena, MediaPipe hand landmarks, deterministic hand-shape rules, FastAPI/MongoDB progress and notification services, and an ESP32/SSD1306/vibration prototype. The design responds to an important usability problem: a learner's hand action can generate an immediate physical cue without requiring the learner to interpret another screen message.

The research value of the prototype lies in its integration and evaluation opportunity rather than in a claim of state-of-the-art recognition. The current code audit makes this distinction especially important. Several implementation details - including hard-coded UI accuracy, client-trusted pass fields, fixed device status values, and the separation between frontend BLE code and Wi-Fi firmware - must be disclosed. Doing so strengthens the thesis because the artifact can be assessed against evidence rather than marketing claims.

The current report reaches an artifact-level conclusion: the prototype is suitable for controlled evaluation of visual-haptic feedback, but educational effectiveness remains an open empirical question. No participant result is inferred from interface defaults or source-code constants.

8.2 Final Research Statement

The individual contribution is a low-cost, explainable visual-haptic practice artifact and a reproducible measurement framework. Future work should unify the communication stack, replace finger-count heuristics with validated spatiotemporal recognition, calibrate haptic intensity, independently validate sign definitions with qualified sign-language experts, and evaluate the system with the intended Deaf and hard-of-hearing learner population.

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